ViralFuse — Multimodal Immune Evasion Forecaster

PhD Proposal Prototype · Proof of Concept

Overview

ViralFuse is a multimodal deep learning prototype for forecasting viral immune evasion in SARS-CoV-2 spike protein variants. It integrates four complementary data modalities through a cross-attention transformer fusion architecture.

Modalities

#	Modality	Source	Dim	Status
1	Protein Sequence	ESM-2 (8M)	320	✅ Real
2	3D Structure	AlphaFold2 (pLDDT, contact map, SASA)	64	🟡 Simulated
3	MD Simulation	GROMACS (RMSD, RMSF, ΔΔG, H-bonds)	32	🟡 Simulated
4	Epitope Vector	CoV-AbDab classes I–IV	16	🟡 Rule-based

Architecture

```
[Seq] [Struct] [MD] [Epitope]
  ↓      ↓      ↓      ↓
 Linear projections → shared 64-dim space
          ↓
   Cross-Attention Fusion
          ↓
   Pooled representation
          ↓
   Ridge Regression Head
          ↓
   Immune Escape Score ∈ [0, 1]
```

Quick Start

```bash
pip install -r requirements.txt
streamlit run app.py
```

Features

- **Single Mutation** — predict escape score with modality attribution breakdown
- **Variant Trajectory** — forecast immune evasion across an ordered mutation chain
- **Batch Prediction** — run predictions across many mutations, export CSV
- **Architecture Tab** — full model explanation with references

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Full Pipeline (Research Extension)

Replace simulated modalities with:

- **ESM-2 650M** for richer sequence embeddings
- **AlphaFold2 full inference** → extract per-residue pLDDT, contact maps, SASA
- **100ns GROMACS MD** on spike–ACE2 and spike–antibody complexes → MM-PBSA $\Delta\Delta G$
- **CoV-AbDab** for epitope class annotation
- **End-to-end training** on DMS escape datasets (Bloom 2022, Cao 2022, Starr 2022)
- **MLP head** with uncertainty quantification (MC Dropout)

Key References

- Lin et al. (2022) — ESM-2 · *Nature*
- Jumper et al. (2021) — AlphaFold2 · *Nature*
- Bloom et al. (2022) — DMS escape mapping · *Cell Host & Microbe*
- Cao et al. (2022) — Omicron evasion landscape · *Nature*
- Starr et al. (2022) — RBD DMS · *PLOS Pathogens*
- Hie et al. (2021) — Evolutionary velocity · *Science*